REVIEW REPORT

Title: On topological characterizations of rough automata and their primary decomposition
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This work develops a topology-driven approach to rough automata by defining definite and possible core operators and using them to generate natural saturated topologies on the state set. The paper connects genetic closedness and primary subautomata to regular closed sets, and extends these correspondences to decomposition results for compact rough automata, including finiteness of primaries and primary decomposition theorems. Product behavior is also examined through categorical constructions and induced product topologies. The topic is relevant to the interface of rough set theory, topology, and automata, and the results appear potentially useful for further categorical and decomposition studies. The paper may be considered for publication after major revision.

1. On page 4: ("Thought this section" → **Throughout this section**)
2. On page 8, Proposition 3.5: ("precisely a a minimal" → **precisely a minimal**)
3. On page 10, Definition 4.1: ("(Q, X, δ)" → $(\boldsymbol{Q, \sim, X, \delta})$)
4. On page 10, Lemma 4.1: ("FuA" → **RAut**)
5. On page 11, Proposition 4.4: ("$M = (Q, R, X, \delta)$" → $M = (\boldsymbol{Q, \sim, X, \delta})$)
6. There are some sentences which have unclear meaning in the abstract, authors should correct them.
7. Discuss the limitations of existing studies with the proposed one.
8. Authors should use common symbols which can be easily understood and read.
9. Cite all references in the text.
10. Irrelevant references should be deleted from the list of references.